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Apparatus and method for heat source localization and peak temperature estimation

Abstract

A heat source sensor includes a plurality of first wires extending in a first direction, a plurality of second wires extending in a second direction different from the first direction and crossing the plurality of first wires, and a plurality of nodes. Each node is defined at a crossing of a first wire of the plurality of first wires and a second wire of the plurality of second wires. The first wire is secured to the second wire at the node. Each node of the plurality of nodes defines a measurement point of the heat source sensor, with a difference in thermoelectric electromotive forces between two nodes of the plurality of nodes indicative of a temperature difference between the two nodes.

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Field of Classification Search**CPC:** G06N (3/0635); G06N (3/08); G06F (17/11); H01L (35/32); H01L (35/34)**References Cited****U.S. PATENT DOCUMENTS**

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Primary Examiner: Cole; Brandon S*Attorney, Agent or Firm:* CANTOR COLBURN LLP**Background/Summary**

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 62/714,991, filed Aug. 6, 2018, which is incorporated herein by reference in its entirety.

BACKGROUND

(1) Exemplary embodiments pertain to the art of heat source location identification and strength estimation.

(2) Central to a wide range of challenging engineering problems is the need for fast and accurate identification of the location and strength of a small heat source, whether the heat source is induced or spontaneous, over a large area. Induced heat sources are ubiquitous in manufacturing processes of thermo-mechanical origin, such as laser processing, machining, and welding, where location and strength of heat sources are important process parameters that need to be monitored in real-time for quality assurance, diagnose and prognosis. Vibration induced heat generation is used to locate cracks in materials for non-destructive testing where locations of heat sources need to be identified as accurately as possible.

(3) On the other hand, spontaneous heat sources are often signs of the onset of critical changes in a system that require immediate attention, such as thermal runaway in lithium ion batteries (LIBs), short circuits in integrated circuits (ICs) or solar cells, localized rise of body temperatures due to tumors, or abrupt temperature change of the surface of a hypersonic vehicle due to boundary layer transition from laminar to turbulent, etc.

(4) Direct experimental localization and strength estimation of heat sources requires measurement systems with not only high spatial and temporal resolution but also high Spatial Dynamic Range (SDR), which is so far achievable only by infrared (IR) thermal imaging techniques made possible by a wave sensing device such as a Focal Plane Array (FPA) thanks to the advancements in microsystem technologies and the non-intrusive nature of radiation heat transfer. High SDR thermal imaging techniques based on conductive heat transfer, however, is still in its infancy in spite of the pressing needs in situations where even IR cameras fall short (e.g. limited space, harsh environment, internal heat source, etc.), which motivates the studies of Inverse Heat Conduction Problems (IHCP) that seek to numerically locate heat sources from limited sensor inputs. State-of-the-art conductive thermal imaging techniques employ arrays of thermocouples or resistance temperature detectors

(RTDs), which suffer from either high manufacturing cost or low temporal resolution.

BRIEF DESCRIPTION

- (5) In one embodiment, a heat source sensor includes a plurality of first wires extending in a first direction, a plurality of second wires extending in a second direction different from the first direction and crossing the plurality of first wires, and a plurality of nodes. Each node is defined at a crossing of a first wire of the plurality of first wires and a second wire of the plurality of second wires. The first wire is secured to the second wire at the node. Each node of the plurality of nodes defines a measurement point of the heat source sensor, with a difference in thermoelectric electromotive forces between two nodes of the plurality of nodes indicative of a temperature difference between the two nodes.
- (6) Additionally or alternatively, in this or other embodiments the plurality of first wires is perpendicular or at an angle to the plurality of second wires.
- (7) Additionally or alternatively, in this or other embodiments the plurality of first wires are formed from a first material and the plurality of second wires are formed from a second material different from the first material.
- (8) Additionally or alternatively, in this or other embodiments the plurality of first wires are Chromel®, and the plurality of second wires are Alumel®.
- (9) Additionally or alternatively, in this or other embodiments the first wire is welded to the second wire at each node.
- (10) Additionally or alternatively, in this or other embodiments the plurality of first wires and the plurality of second wires are arranged in a woven configuration.
- (11) In another embodiment, a heat source localization and peak temperature estimation system includes a heat source sensor including a plurality of first wires extending in a first direction, a plurality of second wires extending in a second direction different from the first direction and crossing the plurality of first wires, and a plurality of nodes. Each node is defined at a crossing of a first wire of the plurality of first wires and a second wire of the plurality of second wires. The first wire is secured to the second wire at the node. Each node of the plurality of nodes defines a measurement point of the heat source sensor, with a difference in thermoelectric electromotive forces between two nodes of the plurality of nodes indicative of a temperature difference between the two nodes. A data acquisition system is configured to receive the plurality of thermoelectric electromotive forces from the plurality of nodes, compare the plurality of thermoelectric electromotive forces to a mathematical model, and determine a heat source location and peak temperature based on a result of the comparison.
- (12) Additionally or alternatively, in this or other embodiments the plurality of first wires is perpendicular to the plurality of second wires.
- (13) Additionally or alternatively, in this or other embodiments the plurality of first wires are formed from a first material and the plurality of second wires are formed from a second material different from the first material.
- (14) Additionally or alternatively, in this or other embodiments the plurality of first wires are Chromel®, and the plurality of second wires are Alumel®.
- (15) Additionally or alternatively, in this or other embodiments the first wire is welded to the second wire at each node.
- (16) Additionally or alternatively, in this or other embodiments the plurality of first wires and the plurality of second wires are arranged in a woven configuration.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:
- (2) FIG. 1 is a perspective view of an embodiment of a heat source sensor;
- (3) FIG. 2 is a schematic illustration of an equivalent circuit defined by an embodiment of a heat source sensor;
- (4) FIG. 3 is a schematic illustration of a test apparatus for a heat source sensor;
- (5) FIG. 4 is a schematic illustration of a method of machine learning model training and validation;
- (6) FIG. 5 is a schematic illustration of an ANN model;
- (7) FIG. 6 is a schematic illustration of an LSTM model;
- (8) FIG. 7 is a schematic illustration of a hidden unit; and
- (9) FIGS. 8a-8g are a schematic illustration of a method of manufacturing a heat source sensor.

DETAILED DESCRIPTION

- (10) A detailed description of one or more embodiments of the disclosed apparatus and method are presented

herein by way of exemplification and not limitation with reference to the Figures.

(11) The present disclosure relates to a new thermal conduction based heat source sensing mechanism realized by thermocouple wires assembled in a mesh configuration (Thermomesh), whose characteristics are described by a linear model of a finite grid of voltage sources with internal resistances. The source localization and strength estimation problem is then translated to a mathematical one which is to solve an ill-posed underdetermined linear system. Thermomesh is different from all existing sensors (including contact and non-contact) for heat source localization and strength estimation purposes in that the inverse of all existing sensor models are always well-posed with unique and stably solvable solutions but the inverse of the sensor model of Thermomesh is ill-posed with no unique and stable solution without prior information of the heat source. While numerous solution techniques are available to solve IHCP with measurements from traditional sensors, the possibility of solving IHCP using the Thermomesh sensor is unknown. Here, we demonstrate the potential of Machine Learning (ML) algorithms toward this goal. Source localization and strength estimation are respectively achieved with high confidence by Deep Learning (DL) based classification and Artificial Neural Network (ANN) based prediction algorithms without the need to compute the solution, i.e. the temperature distribution.

(12) Referring now to FIG. 1, a thermomesh sensor **10** includes a plurality of thermocouple wires **12** assembled (or welded, printed, deposited, etc.) in a wire mesh configuration. The plurality of thermocouple wires **12** includes a plurality of first wires **12a** extending in a first direction, and a plurality of second wires **12b** extending in a second direction. A node **14** is defined where a first wire **12a** crosses a second wire **12b**, with the plurality of first wires **12a** and the plurality of second wires **12b** defining a plurality of nodes **14**.

(13) In some embodiments, the plurality of first wires **12a** are parallel-extending, and the plurality of second wires **12b** are parallel-extending. Further, in some embodiments, each first wire **12a** of the plurality of first wires **12a** is perpendicular or at an angle to each second wire **12b** of the plurality of second wires **12b**. One skilled in the art will readily appreciate that this configuration is merely exemplary, and other configurations of thermocouple wires **12** to define the plurality of nodes **14** are contemplated within the scope of the present disclosure. In the embodiment illustrated in FIG. 1, the first wires **12a** and the second wires **12b** are arranged in a woven arrangement and secured at each node **14**. In some embodiments, the plurality of first wires **12a** are Chromel®, while the plurality of second wires **12b** are Alumel®. In some embodiments, the plurality of thermocouple wires **12** are arrayed on or in a substrate **20**. Further, in some embodiments the nodes **14** are arranged approximately 10 μm apart, but in other embodiments other node spacing may be utilized.

(14) Each node **14** or junction is electrically connected, as shown in FIG. 1 and represents a heat source sensing point, which is in thermal equilibrium with the measurand (schematically illustrated at 16) in contact with the thermomesh sensor **10**, or object to be measured. Temperature differences between adjacent nodes **14** generate thermoelectric electromotive forces (EMF) in the thermocouple wires due to the Seebeck effect. Wire segments **18** connecting adjacent nodes **14** are therefore be considered as voltage sources with an internal resistance, whose EMF is determined by the temperature difference and a Seebeck coefficient of the wire segments **18** between adjacent nodes **14**, leading to an equivalent circuit of a M×N Thermomesh sensor shown in FIG. 2.

(15) Applying Kirchhoff current law, the net current entering each node **14** is equal to zero and a system of linear equations is obtained and shown as Equation 1 below:

$$\begin{aligned} &C_{sub.i;j;i+1;j}(U_{sub.i;j}-U_{sub.i+1;j}+s_{sub.n}T_{sub.i;j}-s_{sub.n}T_{sub.i+1;j}) \\ &+C_{sub.i;j;j-1;j}(U_{sub.i;j}-U_{sub.i-1;j}+s_{sub.n}T_{sub.i;j}-s_{sub.n}T_{sub.i-1;j}) \\ &+C_{sub.i;j;j;j+1}(U_{sub.i;j}-U_{sub.i;j+1}+s_{sub.p}T_{sub.i;j}-s_{sub.p}T_{sub.i;j+1}) \\ &+C_{sub.i;j;j;j-1}(U_{sub.i;j}-U_{sub.i;j-1}+s_{sub.p}T_{sub.i;j}-s_{sub.p}T_{sub.i;j-1})=0; \end{aligned} \quad (1)$$

(16) where $C_{sub.i;j;k;1}$ is the wire conductance between node (i; j) and node (k; 1), $s_{sub.n}$ the negative Seebeck coefficient of the first wires **12a**, $s_{sub.p}$ the positive Seebeck coefficient of the second wires **12b**, $U_{sub.i;j}$ the electric potential of node (i; j), and $T_{sub.i;j}$ the difference between temperature of node (i; j) and the cold-junction where the exterior boundary potentials $V=[V_1, V_2, \dots, V_{2M+2N}]$ are measured. An additional precision analog silicon temperature sensor **20**, such as (LM34CAZ from Texas Instruments) measures the cold junction temperature.

(17) After rewriting Equation 1 in matrix form, Equation 2, below, is obtained.

$$(18) \quad CU = C \begin{bmatrix} U_{1,1} \\ U_{1,2} \\ \vdots \\ U_{M,N} \end{bmatrix} = S \begin{bmatrix} T_{1,1} \\ T_{1,2} \\ \vdots \\ T_{M,N} \end{bmatrix} = ST \quad (2)$$

(19) where $U=[U_{1,1}; U_{1,2}; \dots; U_{M,N}]$ is the vectorized potential matrix for internal nodes, C the conductance matrix, S the Seebeck coefficient matrix, and T the vectorized temperature matrix. The exterior boundary potentials $V=[V_1; V_2; \dots; V_{2M+2N}]$ to be measured, are related to vector T by

$$V=AT; \quad (3)$$

(20) where A is a $2M+2N$ by MN matrix.

(21) The complete reconstruction of T from measured V is therefore an ill posed underdetermined inverse problem. However, we show strong evidence of accurate heat source localization and peak temperature estimation via numerical experiments.

(22) Referring now to FIG. 3, an experiment was designed to verify the physical model on a 3×3 Thermomesh **30** using bare K-type thermocouple wires resistively welded together at each of the 9 junctions **32** and covered from both sides by polyimide tapes as shown in FIG. 3. For each measurement, a CO.sub.2 laser beam **34** approximately 3 mm wide is focused on one of the junctions **32** for an extended period of time allowing the temperatures to be recorded on the thermal camera **36** to reach steady state. A total of 27 thermal images were taken with laser focused sequentially on each of the 9 junctions **32** at three different power levels 36 W, 40 W, and 44 W, while the 12 exterior boundary potentials being streamed at 0.5 s per cycle for approximately 10 s by a multichannel data acquisition system **38**. The 9 junction temperatures on each thermal image are then extracted and plugged into the model to calculate the exterior boundary potential vector V and compared with the experimentally measured ones.

(23) The average error between calculated and experimental potentials is 54 μ V, corresponding to less than 2 degrees Celsius for K-type thermocouples which has a sensitivity of approximately 41.6 μ V per degree Celsius (32.8 μ V per degree Celsius for Chromel and -8.8 μ V per degree Celsius for Alumel at 900 degrees Celsius) suggesting agreement between the model and sensor response to the given temperature inputs.

(24) Successful validation of the physical model allowed for performance of numerical experiments that reliably predict the experimental outcome at much higher data generation rate and reduced costs. By letting the Thermomesh sensors **10** have negligible thermal mass compared to the measurand, which can be realized by using thinner thermocouple wires, the changes in the exterior boundary potentials follow the change of temperature inputs almost instantaneously, permitting in-situ temperature measurement of which the temporal resolution is only limited by the sampling rate of the data acquisition device **36** and the computational time. For the purpose of demonstrating the capability of the Thermomesh sensors **10** for high spatial resolution heat source localization and peak temperature estimation, a two-dimensional Gaussian function was utilized with randomly generated locations, standard deviations, and peak values as temperature inputs to 1) calculate the exterior boundary potentials and 2) train the Machine Learning (ML) models. A deep learning (DL) based classification is used for heat source localization and an artificial neural network (ANN) based prediction is used for peak temperature estimation.

(25) A flowchart depicting the procedures for ML model training and validation is shown in FIG. 4. The experiment is considered a success if the trained ML model is able to accurately identify the location and the peak value of the temperature inputs with high confidence, when given the calculated potentials as inputs. A K-type 10×10 Thermomesh sensor **10** is used in the numerical experiments.

(26) The peak temperature of a measurement is predicted with an artificial neural network (ANN) based model, as shown in FIG. 5. The applied ANN structure includes an input layer **40**, one hidden layer **42**, and an output layer **44**. The N input data $x=\{x_{1,1}; x_{1,2}; \dots; x_{M,N}\}$ is fed to the input layer **40**. The hidden layer **42** includes multiple (e.g. M) neurons. Each neuron, e.g. $h_{j,1}$, is a function represented as equation 3, below

$$h_{j,1} = \sigma(\sum_{i=1}^N w_{i,1} x_{i,1} + b_{1,1}) \quad (3)$$

where $\sigma(*)$ is an activation function as follows:

$$\sigma(t) = 1/(1 + e^{-t}) \quad (4)$$

(27) The activation function provides a smooth transition as input values change. $W_{sup}(1)=[w_{1,1,1}; w_{1,2,1}; \dots; w_{M,N,1}]$ is a weight matrix and $b_{sup}(1)=[b_{1,1,1}; b_{1,2,1}; \dots; b_{M,N,1}]$ is a bias vector. The output is computed from a regression layer as shown in Equation 5:

$$(28) \hat{y} = \sum_{i=1}^M \text{Math.} (w_i^{(2)} h_i + b_i^{(2)}) \quad (5)$$

(29) where $W_{\text{sup.}(2)} = [w_{\text{sub.1.sup.}(2)}, w_{\text{sub.2.sup.}(2)}, \dots, w_{\text{sub.N.sup.}(2)}]$ is a weight matrix and $b_{\text{sup.}(2)} = [b_{\text{sub.1.sup.}(2)}, b_{\text{sub.2.sup.}(2)}, \dots, b_{\text{sub.N.sup.}(2)}]$ is a bias vector. The ANN structure includes one input layer 40, one hidden layer 42 and one output layer 44. The hidden layer includes 200 neurons. Minimum square error (MSE) is applied as a loss function, defined as in Equation 6:

$$(30) \begin{aligned} L &= \frac{1}{2} \sum_{k=1}^K (\mathbf{y}^k - \hat{\mathbf{y}}^k)^2 \\ &= \frac{1}{2} \sum_{k=1}^K (\mathbf{y}^k - (\mathbf{w}^{(2)} \mathbf{h}^k + \mathbf{b}^{(2)}))^2 \\ &= \frac{1}{2} \sum_{k=1}^K (\mathbf{y}^k - (\mathbf{w}^{(2)} (\mathbf{w}^{(1)} \mathbf{x}^k + \mathbf{b}^{(1)}) + \mathbf{b}^{(2)}))^2 \end{aligned} \quad (6)$$

(31) where K is the total number of samples, $y_{\text{sup.k}}$ is the actual value of the k-th sample $x_{\text{sup.k}}$. $\epsilon_{\text{sub.k}} = (y_{\text{sub.k}} - \hat{y}_{\text{sup.k}})$. The weights and biases are updated through back-propagation by computing a gradient of each of the values as shown below in Equations 7-10:

$$(32) \frac{\partial L}{\partial w_i^{(2)}} = - \epsilon_{\text{sub.k}} h_i^k, \quad (7) \quad \frac{\partial L}{\partial b_i^{(2)}} = - \epsilon_{\text{sub.k}}, \quad (8) \quad \frac{\partial L}{\partial w_i^{(1)}} = - \epsilon_{\text{sub.k}} w_i^{(2)} (\mathbf{w}^{(1)} \mathbf{x}^k + \mathbf{b}^{(1)}) x_i^k, \quad (9)$$

$$\frac{\partial L}{\partial b_i^{(1)}} = - \epsilon_{\text{sub.k}} w_i^{(2)} (\mathbf{w}^{(1)} \mathbf{x}^k + \mathbf{b}^{(1)}), \quad (10)$$

(33) The weights and biases are then updated as follows in Equation 11 and 12:

$$(34) \begin{aligned} w_i^{(2)} &\leftarrow w_i^{(2)} - \frac{\partial L}{\partial w_i^{(2)}}, & w_i^{(1)} &\leftarrow w_i^{(1)} - \frac{\partial L}{\partial w_i^{(1)}}, \\ b_i^{(2)} &\leftarrow b_i^{(2)} - \frac{\partial L}{\partial b_i^{(2)}}, & b_i^{(1)} &\leftarrow b_i^{(1)} - \frac{\partial L}{\partial b_i^{(1)}}. \end{aligned} \quad (11) \quad (12)$$

(35) The location of the highest temperature in a measurement is estimated with a neural network model that is based on a modified long short term memory (LSTM) network. The focus in this area is on a single measurement. Therefore, the time step is eliminated from the original network configuration. If a prediction shall be made from multiple measurements, the memory can be easily added back to the network configurations. Once the location of the highest temperature measurement, or location of the heat source is obtained, the temperature or heat flux of distributed points is calculated from the measured heat source information.

(36) An overview of the location prediction scheme is shown in FIG. 6. Each entry into the input layer 40 is a vector that includes N voltage measurements, i.e. $\mathbf{x} = \{x_{\text{sub.1}}, x_{\text{sub.2}}, \dots, x_{\text{sub.N}}\}$. The hidden layer 42 includes M hidden units that are modified from an LSTM unit. The output layer 44 includes a Softmax and classification layer that outputs the results to a class of locations.

(37) FIG. 7 illustrates a structure of a first hidden unit, with the remaining hidden units having identical structures. As shown, the output of each hidden unit is generated from three functions, g, i, and o. Assuming the output vector is $x_{\text{sub.j}}$, the t-th hidden unit outputs as shown in Equation 13 below:

$$\begin{aligned} h_{\text{sub.t}} &= \tanh(i_{\text{sub.t}} \odot g_{\text{sub.t}}) \odot o_{\text{sub.t}} \\ &= \tanh(\sigma(w_{\text{sub.gx.sub.j}} + b_{\text{sub.i}}) \odot \tanh(w_{\text{sub.gx.sub.j}} + v_{\text{sub.g}})) \odot \sigma(w_{\text{sub.ox.sub.i}} + b_{\text{sub.o}}). \end{aligned} \quad (13)$$

(38) where $\odot$ indicates the Hadamard product; $w_{\text{sub.g}}, w_{\text{sub.i}}, w_{\text{sub.o}}$ are the weights applied to functions g, i, and o respectively; and $b_{\text{sub.g}}, b_{\text{sub.i}}, b_{\text{sub.o}}$ are the biases applied to the functions g, i, and o respectively. The weights and biases are updated using back propagation that is similar to the ANN model. Note that the time step is not applied in the proposed model, thus back propagation is used, as opposed to back propagation through time.

(39) The conductance matrix C is a $MN \times MN$ matrix in which each row contains a maximum of five non-zero elements that are coefficients of relevant node potentials in equation 1. The number of non-zero elements is reduced to four for internal boundary nodes, and three for internal corner nodes because no current flows between internal boundary nodes and exterior boundary nodes. The resistivity of horizontal wires is denoted with a positive Seebeck coefficient as $p_{\text{sub.p}}$, and the resistivity of vertical wires is denoted with a negative Seebeck coefficient as $p_{\text{sub.n}}$. Horizontal distances between adjacent nodes is $d_{\text{sub.p}}$, and vertical distances between adjacent nodes is $d_{\text{sub.n}}$. If the cross-sectional area of all wires are assumed to be identical, the potentials are then independent of the wire diameters which therefore do not appear in matrix C. For a 3×3 Thermomesh sensor, the conductance matrix C takes the form:

$$(40)$$

$$\begin{bmatrix}
R_p + R_n & -R_p & 0 & -R_n & 0 & 0 & 0 & 0 & 0 \\
-R_p & 2R_p + R_n & -R_p & -R_n & 0 & 0 & 0 & 0 & 0 \\
0 & -R_p & R_p + R_n & 0 & 0 & -R_n & 0 & 0 & 0 \\
-R_n & 0 & 0 & R_p + 2R_n & -R_p & 0 & -R_n & 0 & 0 \\
0 & -R_n & 0 & -R_p & R_p + 2R_n & -R_p & 0 & -R_n & 0 \\
0 & 0 & -R_n & 0 & -R_p & R_p + 2R_n & 0 & 0 & -R_n \\
0 & 0 & 0 & -R_n & 0 & 0 & R_p + R_n & -R_p & 0 \\
0 & 0 & 0 & 0 & -R_n & 0 & -R_p & 2R_p + R_n & -R_p \\
0 & 0 & 0 & 0 & 0 & -R_n & 0 & -R_p & R_p + R_n
\end{bmatrix}, \quad (14)$$

where

$$(41) R_p = \frac{d_p}{p}, \text{ and } R_n = \frac{d_n}{n}.$$

(42) The corresponding Seebeck coefficient matrix S takes the form:

$$\begin{bmatrix}
-S_p - S_n & -S_p & 0 & S_n & 0 & 0 & 0 & 0 & 0 \\
S_p & -2S_p - S_n & S_p & S_n & 0 & 0 & 0 & 0 & 0 \\
0 & S_p & -S_p - S_n & 0 & 0 & S_n & 0 & 0 & 0 \\
S_n & 0 & 0 & -S_p - 2S_n & S_p & 0 & S_n & 0 & 0 \\
0 & S_n & 0 & S_p & -2S_p - 2S_n & S_p & 0 & S_n & 0 \\
0 & 0 & S_n & 0 & S_p & -S_p - 2S_n & 0 & 0 & S_n \\
0 & 0 & 0 & S_n & 0 & 0 & -S_p - S_n & S_p & 0 \\
0 & 0 & 0 & 0 & S_n & 0 & S_p & -2S_p - S_n & S_p \\
0 & 0 & 0 & 0 & 0 & S_n & 0 & S_p & -S_p - S_n
\end{bmatrix}, \quad (15)$$

where

$$(44) S_p = \frac{s_p d_p}{p}, \text{ and } S_n = \frac{s_n d_n}{n}.$$

(45) Since C is a singular matrix, the first column of C may be eliminated by setting U.sub.1.1=0, so a unique solution exists as

$$U = C \cdot \text{sup.} - 1ST, \quad (16)$$

(46) The exterior boundary potentials may then be calculated as shown in Equation 17 as:

$$\begin{aligned}
& \begin{matrix} V_1 & U_1 & T_{1,1} & 0 \\ V_2 & U_2 & T_{1,2} & 0 \\ \text{.Math.} & \text{.Math.} & \text{.Math.} & \text{.Math.} \\ V_N & U_N & T_{1,N} & 0 \\ V_{N+1} & U_{N+1} & 0 & T_{1,N} \\ V_{N+2} & U_{N+2} & 0 & T_{2,N} \\ \text{.Math.} & \text{.Math.} & \text{.Math.} & \text{.Math.} \\ V_{N+M} & U_{N+M} & 0 & T_{M,N} \\ V_{N+M+1} & U_{N+M+1} & T_{M,N} & 0 \\ V_{N+M+2} & U_{N+M+2} & T_{M,N-1} & 0 \\ \text{.Math.} & \text{.Math.} & \text{.Math.} & \text{.Math.} \\ V_{2N+M} & U_{2N+M} & T_{M,1} & 0 \\ V_{2N+M+1} & U_{2N+M+1} & 0 & T_{M,1} \\ V_{2N+M+2} & U_{2N+M+2} & 0 & T_{M-1,1} \\ \text{.Math.} & \text{.Math.} & \text{.Math.} & \text{.Math.} \\ V_{2N+2M} & U_{2N+2M} & 0 & T_{1,1} \end{matrix} \\
(47) \quad 0 \quad V = [&] = [&] + s_n [&] + s_p [&] = U_{2M+2N} + T_{2M+2N}. \quad (17)
\end{aligned}$$

(48) Finally, all internal and exterior potentials are subtracted by V .sub.1 so that V .sub.1=0 to be consistent

with the measurement and rearranged to obtain:

$$v=(I_{\text{sub.UC}}\sup.-1S+I_{\text{sub.S}})T=AT, \text{ where } I_{\text{sub.U}}=U_{\text{sub.2M}}+2NU_{\text{sup.}}-1 \text{ and } I_{\text{sub.S}}=T_{\text{sub.2M}}+2NT_{\text{sup.}}-1 \quad (18).$$

(49) Referring now to FIGS. **8a-8g**, illustrated is a schematic view of a method of fabricating a thermomesh sensor **10**. In FIG. **8a**, the substrate **20** is coated with a resist layer **80** and exposed to UV light **82** through a photomask **84**. In FIG. **8b**, the resist layer **82** is developed, resulting in an undercut **86**. A wire material **12b**, for example, Alumel, is applied via, for example, sputtering. The remaining resist layer **82** is then removed via, for example, acetone or other material, leaving the wire **12b** material at the substrate **20** as shown in FIG. **8d**. The steps in **8a** through **8c** are repeated for the first wire **12a** material, for example, Chromel, with the result illustrated in FIG. **8e**. In some embodiments, the deposition of the second wire **12b** material is thicker than the deposition of the first wire **12a** material to ensure continuity at the nodes **14**. In FIG. **8f**, the remaining resist layer **82** is removed, leaving the wire **12a** material. In some embodiment, as shown in FIG. **8g**, an insulation layer **88** is applied over the wire **12a** material and the wire **12b** material. The insulation layer **88** may be formed from, for example, SiO₂.

(50) The term “about” is intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application.

(51) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

(52) While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

Claims

1. A heat source sensor, comprising: a plurality of first wires extending in a first direction; a plurality of second wires extending in a second direction different from the first direction and crossing the plurality of first wires, the plurality of first wires and the plurality of second wires arranged in a wire mesh configuration; a plurality of nodes, each node defined by a direct connection of a first wire of the plurality of first wires to a second wire of the plurality of second wires; wherein each node of the plurality of nodes defines a measurement point of the heat source sensor, with a difference in thermoelectric electromotive forces between two nodes of the plurality of nodes is utilized to calculate a temperature difference between the two nodes, the temperature difference is utilized to identify a location of a heat source; and wherein the first wire is welded to the second wire at each node.
 2. The heat source sensor of claim 1, wherein the plurality of first wires is perpendicular or at an angle to the plurality of second wires.
 3. The heat source sensor of claim 1, wherein the plurality of first wires are formed from a first material and the plurality of second wires are formed from a second material different from the first material.
 4. The heat source sensor of claim 1, wherein the plurality of first wires are Chromel®, and the plurality of second wires are Alumel®.
 5. The heat source sensor of claim 1, wherein the plurality of first wires and the plurality of second wires are arranged in a woven configuration.
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